

LogLens - Project Report

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Project Overview

LogLens is an AI-powered log analysis REST API built with Perl 5 and the Mojolicious web framework. It provides endpoints for log ingestion, pattern detection, anomaly identification, log search, statistics aggregation, and AI-powered analysis via a MockClaudeClient.

Technical Stack

Language: Perl 5.34.1

Framework: Mojolicious 9.42

Architecture: REST API with in-memory storage

AI Integration: MockClaudeClient (deterministic pattern-based analysis)

Test Framework: Test::More + Test::Mojo

Tests Passing: 183/183

API Endpoints

- GET /health - Health check
- POST /api/logs - Ingest log entries
- GET /api/logs - List/search logs with filtering
- GET /api/logs/:id - Get specific log entry
- DELETE /api/logs/:id - Delete log entry
- POST /api/logs/analyze - AI-powered log analysis
- POST /api/patterns/detect - Detect patterns in logs
- POST /api/anomalies/detect - Detect anomalies
- GET /api/stats - Aggregated statistics
- POST /api/logs/search - Advanced search with filters

Key Features

- Log ingestion with severity levels (DEBUG, INFO, WARN, ERROR, FATAL)
- Pattern detection using regex and frequency analysis
- Anomaly detection based on statistical deviation
- Full-text search with level and timestamp filtering
- MockClaudeClient provides deterministic AI analysis without external API calls

DevOps

Docker: Multi-stage build, perl:5.40-slim base, non-root user

CI/CD: GitHub Actions - test (prove -l t) + Docker build

Repository: <https://github.com/malbinjvc/log-lens>

Errors Encountered

- PERL5LIB path required for Mojolicious (user-local install)
- Uninitialized \$ _ warning in Clients.pm pattern matching
- Docker CPAN module installation for Mojolicious + JSON::XS